CO1 - AT1

Name: C. Manohar

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SIMATS ENGINEERING
ASSESSMENT TOOL

COURSE NAME: **Theory of Computation** **COURSE CODE:** **CSA1301**

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:25**
Duration : 1 Hour

Analytical Problem

Code of Conduct:

I C.Manohar / 192372246 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Manu

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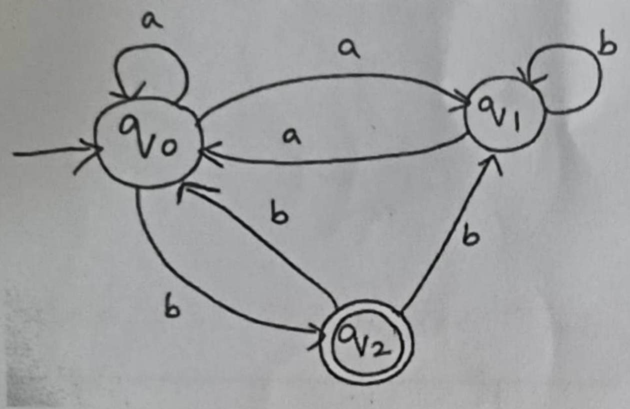
1.Consider the NFA given below which accepts some language L over the alphabet $\Sigma=\{a,b\}$. Construct a DFA that represents the same language L

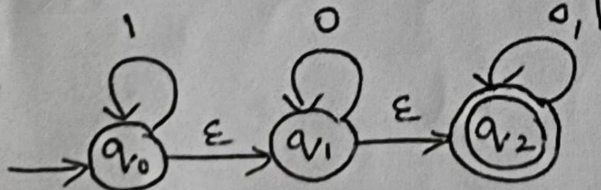


2.Let L be the language of words of length atleast 2 whose last two letters are difference from each other. For example abaab is in L but not bb, Construct an NFA for L, Then draw the equivalent DFA

3. Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string **01011** is accepted by the NFA

4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet $\Sigma=\{0,1\}$. Construct a DFA that represents the same language.

5. A NFA with ε-transitions is given below. Find ε-closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states



Theory of Computation - CSA1301

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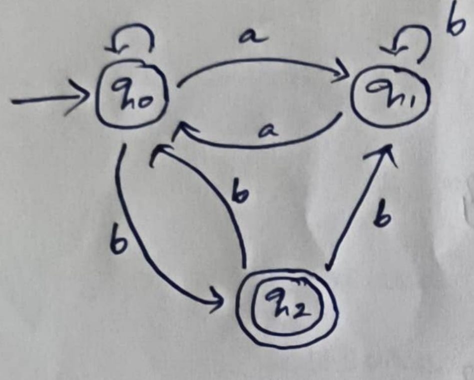
Assessment Tool - 1

① Given state diagram:



Step:2 - Transition Table

state/ip	a	b
q_0	$\{q_0, q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
q_2	$\{\phi\}$	$\{q_0, q_1\}$

Step 3:

Formula: $\{Q, \Sigma, \delta, q_0, F\}$

$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b\}$

$\delta(q_0, a) = \{q_0, q_1\}$

$\delta(q_0, b) = \{q_2\}$

$\delta(q_1, a) = \{q_0\}$

$\delta(q_1, b) : \{q_1\}$

$\delta(q_2, a) = \phi$

$\delta(q_2, b) = \{q_0, q_1\}$

Step 4:

$\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$

$\epsilon\text{-closure}(q_1) = \{q_0, q_1\}$

$\epsilon\text{-closure}(q_2) = \{q_0, q_1, q_2\}$

step 5:

$\delta'(q_0, a) = \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0, a) a))$

$= \epsilon\text{-closure}(\delta(q_0, q_1 q_2), a))$

$= \epsilon\text{-closure}(\delta(q_0, a) \cup \delta(q_1, a) \cup \delta(q_2, a))$

$= \epsilon\text{-closure}(\{q_0, q_1\} \cup \{q_0\} \cup \{\phi\})$

$= \epsilon\text{-closure}(q_1 \cup q_0)$

$= \epsilon\text{-closure}(q_0, q_1) \cup (q_0, q_1, q_2)$

$=$ ~~ε-clos.~~ $\{q_0, q_1, q_2\}$

$\delta'(q_0, b) = \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0), b))$

$= \epsilon\text{-closure}(\delta(q_0, q_1, q_2), b))$

$= \epsilon\text{-closure}(q_2 \cup q_1 \cup \{q_0, q_1\})$

$= \{q_0, q_1, q_2\}$

$$\delta'(q_1,a) = \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_1), a))$$

$$= \epsilon\text{-closure}(\delta(q_0, q_1), a))$$

$$= \epsilon\text{-closure}(\delta(q_0,a) \cup \delta(q_1,a))$$

$$= \epsilon\text{-closure}(\{q_0, q_1\} \cup \{q_0\})$$

$$= \epsilon\text{-closure}(\{q_1 \cup q_0\})$$

$$= \{q_0, q_1, q_2\}$$

$$\delta'(q_1,b) = \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_1), b))$$

$$= \epsilon\text{-closure}(\delta(q_0, q_1), b))$$

$$= \epsilon\text{-closure}(\delta(q_0,b) \cup \delta(q_1,b)$$

$$= \epsilon\text{-closure}(q_2 \cup q_1\}$$

$$= \{q_0, q_1, q_2\}$$

$$\delta'(q_2,a) = \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_2), a))$$

$$= \epsilon\text{-closure}(q_0, q_1 q_2), a))$$

$$= \epsilon\text{-closure}(\delta(q_0,a) \cup \delta(q_1,a) \cup \delta(q_2,a))$$

$$= \epsilon\text{-closure}\{q_0 \cup q_1\}$$

$$= \{q_0, q_1, q_2\}$$

$$\delta'(q_2,b) = \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_2)\ b))$$

$$= \epsilon\text{-closure}(\delta(q_0, q_1 q_2), b))$$

$$= \epsilon\text{-closure}(\delta(q_0,b) \cup \delta(q_1,b) \cup (q_2,b))$$

$$= \epsilon\text{-closure}(q_2 \cup q_1 \cup \{q_0, q_1\})$$

$$= \{q_0, q_1, q_2\}$$

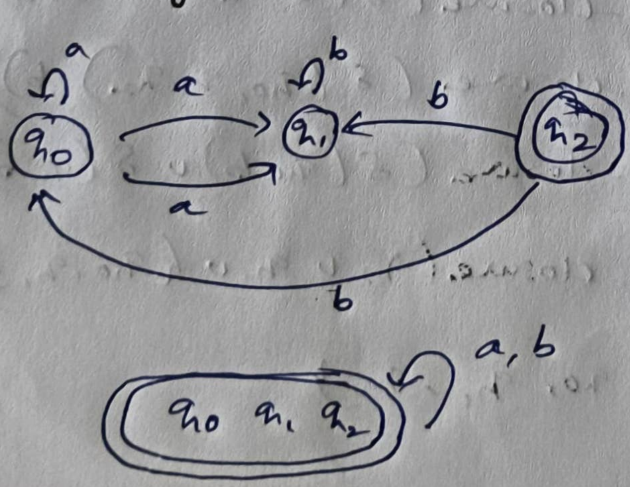
$$\delta'(q_0, q_1, q_2, a) = \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0, q_1, q_2)\ a))$$
$$= \epsilon\text{-closure}(\delta(q_0, a) \cup \delta(q_1, a) \cup \delta(q_2, a))$$
$$= \epsilon\text{-closure}(\{q_0, q_1\} \cup \{q_0\} \cup \phi\}$$
$$= \{q_0, q_1, q_2\}$$

$$\delta'(q_0, q_1, q_2, b) = \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0, q_1, q_2), b))$$
$$= \epsilon\text{-closure}\,\delta(q_0, b) \cup \delta(q_1, b) \cup \delta(q_2, b))$$
$$= \epsilon\text{-closure}\,\delta(q_2 \cup q_1 \cup \{q_0, q_1\})$$
$$= \{q_0, q_1, q_2\}$$

Transition Table:

State/ip	a	b
q_0	$\{q_0, q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
q_2	ϕ	$\{q_0, q_1\}$
$q_0\ q_1\ q_2$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$

NFA State diagram:



(2) Language:

strings of length atleast 2 whose last two symbol are different

Ex:

Accepted	Rejected
ab	aa
ba	bb
aaab	a
abaab	b

NFA:

States

q_0 = starts

q_1 = Last symbol a

q_2 = Last symbol b

q_3 : Final

Transitions:

$q_0 \xrightarrow{a} q_1$

$q_0 \xrightarrow{b} q_2$

$q_1 \xrightarrow{a} q_1$

$q_1 \xrightarrow{b} q_3$

$q_2 \xrightarrow{b} q_2$

$q_2 \xrightarrow{a} q_3$

$q_3 \xrightarrow{a} q_1$

$q_3 \xrightarrow{b} q_2$

Final state:

q_3

Equivalent DFA:

S_0 = Start

S_1 = Ends with a

S_2 = Ends with b

S_3 = Ends with ab (Accept)

S_4 = Ends with ba (Accept)

Transition Table:

State	a	b
S_0	S_1	S_2
S_1	S_1	S_3
S_2	S_4	S_2
S_3	S_4	S_2
S_4	S_1	S_3

Final state:

S_3 S_4

Language:

Ends with

00 (or) 11

Formal defination:

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{0, 1\}$$

$$\text{Start} = q_0$$

$$\text{Final} = \{q_3, q_4\}$$

Transition function:

State	0	1
q_0	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	$\{q_3\}$	ϕ
q_2	ϕ	$\{q_4\}$
q_3	ϕ	ϕ
q_4	ϕ	ϕ

String:

01011

$q_0 \to 0$ $\quad q_0, q_1 \to 1$ $\quad q_0, q_2 \to 0$ $\quad q_0, q_1 \to 1$

$q_0, q_2 \to 1$ $\quad q_4$

Since q_4 is final

$\therefore$ 01011 is Accepted

4\. Language:

$\Sigma = \{0, 1\}$

Need substring:

001

States:

q_0 = No Match

q_1 = Seen 0

q_2 = Seen 00

q_3 = Seen 001 (Final)

Transition Table:

State	0	1
q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_3
q_3	q_3	q_4

Final state:

q_3 this DFA accepts every string containing 001.

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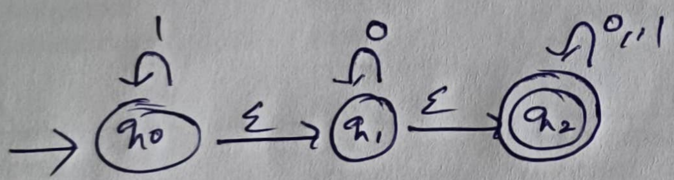
(5)



Given:

q_0 --- ε ---> q_1

q_1 --- ε ---> q_2

q_0 loop on 1

Final:

q_2

$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{0, 1\}$

Start = q_0

Final : $\{q_3, q_4\}$

Transition Function:

State	0	1
A	B	A
B	C	C
C	C	C

DFA state:

$A = \{q_0, q_1, q_2\}$

$B = \{q_1, q_2\}$

$C = \{q_2\}$

Start state:

A

Final state:

A B C

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ASSESSMENT TOOL

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

91/100